Sergei Volkov

Researcher & Scientific Data Engineer · Hydrophysics, Limnology & Data Systems

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Work Authorization: EU Blue Card holder (>2 yrs) · Unrestricted German Work Permit (incl. Freiberufler / Commercial)

Researcher and Scientific Software Engineer bridging empirical physical science (physical limnology, convective turbulence, thermodynamics) and modern high-performance data engineering (lakehouse architectures, distributed data pipelines, Linux systems, and declarative NixOS package maintenance). Co-author of 5 peer-reviewed publications on convective turbulence and thermal stratification in boreal lakes. Extensive industry track record architecting enterprise-scale data infrastructure (Zalando, Playrix, AppIQ) alongside academic research (IGB Berlin, Northern Water Problems Institute).

PROFESSIONAL & RESEARCH EXPERIENCE

Freelance Data / MLOps Engineer | AppIQ
Jan 2026 – Present

Architecture & Implementation of ML-driven Experimentation Platform

- Architected and deployed a self-hosted Lakehouse (**DuckDB/DuckLake on AWS S3**) via **Terraform**, delivering sub-second analytical query latency while eliminating Spark operational overhead.
- Built low-latency ML model serving infrastructure and automated training/evaluation pipelines for real-time model inference.
- Designed high-throughput event ingestion and user tracking pipelines, enabling automated A/B test assignment and real-time metric calculation.
- Standardized declarative infrastructure-as-code and reproducible deployment workflows using **Terraform** and **Nix**.

Sabbatical & Open Source Engineering | Self-Directed
Apr 2025 – Sep 2025

Open Source Contributions & Advanced Technical Upskilling

- Dedicated sabbatical for intensive open-source contributions to **NixOS** (`nixpkgs`) and packages' upstream, packaging and maintaining **Apache Airflow**, **Julia**, and **Databricks CLI**.
- Conducted architectural research on embedded analytical engines (DuckDB) and prepared for advanced machine learning training.

Data Engineer | Zalando SE
Oct 2024 – Mar 2025

Central Data Platform & Analytics Engineering Services

- Developed and maintained business-critical ETL/ELT pipelines using **Managed Workflows for Apache Airflow (MWAA)** and **Databricks Delta Lake**, supporting 15+ internal stakeholder teams.
- Engineered automated CI/CD deployment pipelines for internal Python helper packages and notebook workloads across staging and production Databricks environments.
- Conducted fraudulent customers detection dataset design.
- Provided code reviews, data QA, and architectural support for cross-functional data consumers.

Data Scientist & Cloud Architect | Crocus Labs GmbH
Aug 2023 – Feb 2024

IoT Platform Architecture & Computational Photometry

- Architected scalable cloud infrastructure on **AWS (IoT Core, Lambda, S3, DynamoDB)** for real-time telemetry ingestion from smart greenhouse LED hardware.
- Developed a Python raytracing simulation to model 3D light distribution and optimize LED chip array layouts based on raw photometric emitter data.
- Engineered a containerized IoT test harness with virtualized sensor clients and an **MQTT broker**, enabling end-to-end integration testing without physical hardware.

Data Quality Engineer / Analytics Engineer | Playrix
Nov 2019 – Feb 2023

Promoted across 3 roles (Data Analyst -> QA Engineer -> Data Quality Engineer) for 50M+ DAU platforms

- Led data quality assurance during the migration of legacy Big Data pipelines to a modern lakehouse architecture across **Spark, Hive, Redshift, and Snowflake**.
- Built dynamic SQL metaprogramming validation frameworks that automated data anomaly detection across billions of daily in-game event records.
- Designed and optimized executive **Tableau** dashboards and underlying dimensional data marts tracking player retention, monetization, and A/B test experiments.
- Managed a QA sub-team, establishing automated regression test suites and data verification protocols for engineering releases.

Research Engineer | **Leibniz Institute of Freshwater Ecology (IGB Berlin)** Jun 2021 – Mar 2023
Computer Vision & Environmental Data Infrastructure

- Architected containerized computational environments (**Docker, CMake, MATLAB, OpenCV**) to modernize and execute a legacy 3D particle-tracking codebase.
- Developed computer vision data processing pipelines in **Python** to extract 3D velocity vectors from underwater stereo-video recordings of particulate biomass.
- Administered remote Linux compute nodes and automated data extraction workflows for gigabyte-scale video datasets.

Research Assistant | **Northern Water Problems Institute (NWPI, Russian Academy of Sciences)** Jan 2016 – Nov 2024
Hydrophysical Modeling, Telemetry Analysis & Winter Limnology

- Conducted multi-year observational and numerical research on under-ice convective mixing and boundary-layer heat transfer in boreal lakes.
- Analyzed year-round high-frequency thermistor chain and ADCP telemetry to establish the presence and dynamics of energetic internal waves beneath ice cover.
- Calculated the full Reynolds stress tensor of convective turbulence from paired acoustic Doppler profiler observations.
- Co-authored 5 peer-reviewed publications in international journals (**Environmental Fluid Mechanics**, **Inland Waters**, **Geosciences**).
- Managed full research cycles: observational campaign logistics, sensor calibration, data processing pipelines, and manuscript preparation.

Teaching Assistant | **Petrozavodsk State University** May 2016 – Jul 2020
Department of Energy Supplies & Thermal Engineering

- Delivered lectures, computational seminars, and laboratory coursework in **Thermodynamics**, **Heat and Mass Transfer**, and **Applied Fluid Dynamics**.
- Mentored engineering undergraduate and graduate students in numerical modeling and experimental data analysis.

PEER-REVIEWED JOURNAL PUBLICATIONS

2021 **Full Reynolds Stress Tensor of Convective Turbulence Estimated with Paired Acoustic Doppler Current Profilers**
S. Bogdanov, G. Kirillin, S. Volkov, G. Zdorovennova
Earth and Space Science Open Archive (ESSOAr) · DOI: 10.1002/essoar.10507975.1

2019 **Fine scale structure of convective mixed layer in ice-covered lake**
S. Volkov, S. Bogdanov, R. Zdorovennov, G. Zdorovennova, A. Terzhevik, N. Palshin, D. Bouffard, G. Kirillin
Environmental Fluid Mechanics, 19(5), 1151–1165 · DOI: 10.1007/s10652-018-9652-2

2019 **Under-ice convection dynamics in a boreal lake**
D. Bouffard, G. Zdorovennova, S. Bogdanov, T. Efremova, S. Lavanchy, N. Palshin, A. Terzhevik, L. Vinnå, S. Volkov, A. Wüest, R. Zdorovennov, H. Ulloa
Inland Waters, 9(2), 142–161 · DOI: 10.1080/20442041.2018.1533356

2019 **Structure and dynamics of convective mixing in Lake Onego under ice-covered conditions**
S. Bogdanov, G. Zdorovennova, S. Volkov, R. Zdorovennov, N. Palshin, T. Efremova, A. Terzhevik, D. Bouffard
Inland Waters, 9(2), 162–178 · DOI: 10.1080/20442041.2018.1551655

2018 **Albedo of a Small Ice-Covered Boreal Lake: Daily, Meso-Scale and Interannual Variability on the Background of Regional Climate**
G. Zdorovennova, N. Palshin, T. Efremova, R. Zdorovennov, G. Gavrilenko, S. Volkov, S. Bogdanov, A. Terzhevik
Geosciences, 8(6), 206 · DOI: 10.3390/geosciences8060206

INTERDISCIPLINARY RESEARCH PROJECTS

Lake Pycnoclines Trap Organic Particles Forming Hot Spots of Accelerated Carbon Cycling (PycnoTrap) 2021 – 2023
Leibniz Institute of Freshwater Ecology and Inland Fisheries (IGB Berlin)

Joint limnological project researching sinking velocity, biomass degradation, and hydrodynamic particle trapping across pycnoclines in stratified lake columns. Designed twin-action underwater camera frames and built computer vision / stereo-video tracking pipelines to calculate 3D velocity vectors and estimate vertical particulate carbon flux.

Lake Onego: Life Under the Ice 2015 – 2017
Northern Water Problems Institute (NWPI) / EPFL Limnology Center · Project Website

International multi-institutional field campaigns investigating under-ice convective mixing, thermal boundary regimes, and winter ecosystem dynamics in Europe's second-largest boreal lake.

EDUCATION & ACADEMIC THESES

Specialist Degree in Energy Supplies (M.Sc. Equivalent) 2010 – 2015
Petrozavodsk State University
Diploma Thesis: *"Noniterative Heat Exchanger Calculation"* (Supervisor: Prof. Sergei Bogdanov)
Analytical calculation procedure for heat exchanger sizing, avoiding non-convergent numerical optimization algorithms.

Data Science & Machine Learning Bootcamp Oct 2025 – Dec 2025
Ironhack

Machine Learning Specialization 2017
Stanford University on Coursera

TECHNICAL, SCIENTIFIC & SYSTEMS COMPETENCIES

Domain Physics & Limnology:	Hydrophysics, Physical Limnology, Convective Turbulence, Stratified Flows, Heat & Mass Transfer
Field & Instrumentation:	Acoustic Doppler Current Profilers (ADCP), Thermistor Chains, CTD Profilers, Stereo-Cameras
Data Engineering & Platforms:	Spark, Databricks, DuckDB/DuckLake, Apache Airflow, Snowflake, dbt, Delta Lake, Redshift
Scientific Computing & ML:	Python (NumPy, SciPy, Pandas), MATLAB, Julia, OpenCV, Computer Vision, PyTorch, Scikit-Learn
Systems, DevOps & Infra:	Linux Administration, Docker, Terraform, Nix/NixOS, AWS, CI/CD (GitHub Actions, GitLab CI)
Open Source & Community:	Official Package Maintainer for NixOS (`nixpkgs`: Apache Airflow, Julia, Databricks CLI, plus 6 Python libraries, system utilities) · OpenStreetMap Contributor
Languages:	English (C1/C2 Full Professional), German (B2 Professional Working), Russian (Native)